

Replication Package: Demographic Transition and the Rise of Wealth Inequality

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This replication package accompanies:

Krzysztof Makarski, Joanna Tyrowicz, and Piotr Zoch. (forthcoming). "Demographic transition and the rise of wealth inequality". *Economic Journal*.

Start Here: Replication Routes and Run Order

Most replicators do not need to run every stage. Choose one route first; the detailed steps later in this README use the same numbering and list inputs, outputs, commands, and details separately.

Scope of the Economic Journal Reproducibility Check

The Economic Journal reproducibility team re-downloaded the raw PSID extract from the cited provider-approved repository and verified the baseline `psid_all_govt__` route. The resulting intermediate model outputs agreed with the authors' supplied outputs up to numerical precision (approximately $1e-06$). The team did not independently re-extract the raw ACS/IPUMS data and did not re-solve the remaining 29 model scenarios. For those results, it used the intermediate scenario outputs included in `fortran_code/Results/` and rebuilt the paper figures.

Accordingly, the journal's Data Availability statement will explain that, given the highly demanding algorithms, checks were performed on a subset of results. In this README, Route 3 identifies the fully computationally demanding route, while Route 1 reproduces all figures from the included intermediate outputs.

Route	Goal	Raw restricted data needed?	Run order
1. Figure-only check	Recreate the paper figures from included model output.	No raw PSID or ACS/IPUMS. Requires <code>data/SCF/SCF_plus.dta</code> , <code>data/PSID/psid_ready.dta</code> , and the included <code>fortran_code/Results/</code> folders.	Step 1, then Step 7. Run the MATLAB plot-only command in Step 2B for Appendix Figure B.5 if needed.
2. Rebuild processed calibration inputs	Re-run Stata/MATLAB preprocessing from packaged snapshots, without restricted raw microdata.	No raw PSID or ACS/IPUMS. Packaged processed PSID and ACS-derived files are reused when raw extracts are absent.	Step 1, Step 2A, optional Step 2B (set <code>N_REPS</code> to 0 first for the fast run), then Steps 3-7 only if you also want to re-solve the model.
3. Full raw-data rebuild	Recreate the processed PSID and ACS-derived inputs, re-solve the Fortran model, and regenerate figures.	Yes: <code>inputs_stata_code/income_process/PSID/psid.dta</code> and <code>inputs_stata_code/skill_premium/ACS_college/ACS_college.dta</code> .	Step 1, Step 2A, Step 2B, Step 3, Step 4, Step 6, Step 7.

Route 1: minimal reviewer checklist

1. Confirm `fortran_code/Results/<scenario>` folders are present, especially `psid_all_govt__` and `psid_ndm_govt__`.
2. Confirm `data/SCF/SCF_plus.dta` and `data/PSID/psid_ready.dta` are present.
3. Open `outputs_stata_code/__replication_graphs.stpr` in Stata.
4. Run `outputs_stata_code/__main.do`.
5. Check regenerated figures in `graphs/outputs/` and Appendix-B input figures in `graphs/inputs/`.

Route 3: complete rebuild checklist

1. Place raw PSID at `inputs_stata_code/income_process/PSID/psid.dta` if rebuilding the PSID income-process inputs.
2. Place raw ACS/IPUMS at `inputs_stata_code/skill_premium/ACS_college/ACS_college.dta` if rebuilding the college-share input.
3. Decide whether to keep `global download_data 0` in the Stata drivers or change it to 1 to refresh supported online data sources.
4. Decide the bootstrap setting in `inputs_stata_code/income_process/N_REPS.txt`: the shipped default 1000 reproduces the full paper bootstrap (about 12 hours across Stata and MATLAB); set it to 0 for a fast point-estimate run that reuses the archived paper confidence bands.
5. Run `inputs_stata_code/__main_data_prepare.do` from `inputs_stata_code/main.stpr`. When `psid.dta` is present, this driver runs `I01_run_psid_income_inputs.do` once; do not run it a second time.
6. Run `inputs_matlab_code/income_process/run_income_process_matlab.bat` to estimate the income-process parameters from the Stata outputs.
7. Use the shipped Fortran binaries or rebuild them in Visual Studio, then run the scenarios listed in `fortran_code/scenarios.txt` and the heterogeneous-rate scenarios with `run_scenarios_het.bat`.
8. Run `outputs_stata_code/__main.do` from `outputs_stata_code/__replication_graphs.stpr`.

Manual settings to check before running

Setting	Default	Change only if
global download_data in inputs_stata_code/___main_data_prepare.do and outputs_stata_code/___main.do	0	You want Stata to refresh supported dbnomics/OECD/GGDC series instead of using packaged snapshots.
inputs_stata_code/income_process/N_REP S.txt	1000	You want a fast point-estimate run instead of the full 12-hour paper bootstrap: set it to 0 to reuse the archived 1000-repetition bootstrap files for the confidence bands.
MATLAB_EXE environment variable	C:\Program Files\MATLAB\R2018b\bin\matlab.exe	MATLAB is installed elsewhere or a different version is used.
Raw PSID path	absent by default	You are doing Route 3 and have obtained the PSID extract.
Raw ACS/IPUMS path	absent by default	You are doing Route 3 and have obtained the IPUMS USA extract.

Missing raw PSID or ACS files are not automatically fatal for Routes 1-2. If raw PSID is absent, the Stata driver prints a message and keeps the packaged PSID-derived Fortran inputs and processed `psid_ready.dta`. If raw ACS/IPUMS is absent, the Stata driver keeps the packaged college-share inputs. A full raw-data rebuild requires supplying those raw files at the exact paths listed above.

Replication Package Layout

```

demographic-transition-wealth-inequality/
|-- README.md           This file
|-- LICENSE.txt         MIT license
|-- SCENARIOS.md        Full scenario catalogue
|
|-- fortran_code/        Fortran OLG model (main solver)
|   |-- *.f90           Source files
|   |-- 5Gtrans.sln     Visual Studio solution
|   |-- Data/           Calibration inputs consumed by the model
|   |-- Instructions/   Per-scenario switch files
|   |-- Parameters/     Per-scenario parameter files
|   |-- Results/        Included per-scenario intermediate model output
|   |-- scenarios.txt    Scenario list for batch runs
|   +-- run_scenarios*.bat Batch drivers
|
|-- inputs_stata_code/   Stata code that produces calibration inputs
|   |                   and selected PSID-derived intermediates
|   |-- demography/
|   |-- depreciation/
|   |-- external/        External Fortran inputs (UN/CDC/SSA source data),
|   |                   copied to fortran_code/Data/ via copy_to_fortran.do
|   |-- income_process/ PSID cleaning, variants, omega, covariance txt
|   |   |-- PSID/        Optional raw PSID extract and extraction notes
|   |   |-- I01_0*_.do    Split Stata stages
|   |   +-- I01_run_*.do  PSID income-process Stata launcher
|   |-- labor_share/
|   |-- skill_premium/
|   |-- social_security/
|   |-- tax_rate/
|   +-- tfp/
|
|-- inputs_matlab_code/  MATLAB estimators that consume Stata intermediates
|   +-- income_process/  Income-process parameter estimation
|
|-- outputs_stata_code/  Stata code that produces paper tables/figures
|   |                   from Fortran output and prepared data
|   |-- income_process/  Deaton age-profile plot (M05plot_omega.do)
|   +-- __main.do        Master driver (Step 7 entry point)
|
|-- outputs_matlab_code/ MATLAB plotting code
|   +-- income_process/  Sigma2epsilon plot
|
|-- graphs/             Generated figures (inputs/ and outputs/)
|
+-- data/               Provider data, processed panels, and model-ready panels
                       (SCF+, processed PSID, model panels, etc.; see below)

```

Pipeline at a glance: data/ + inputs_stata_code/income_process/PSID/ -> inputs_stata_code/ + inputs_matlab_code/ -> fortran_code/Data/ -> fortran_code/ (OLG model solver) -> fortran_code/Results/ -> outputs_stata_code/ + outputs_matlab_code/ -> graphs/.

Path convention: prose paths are relative to the repository root. Fortran run examples change into `fortran_code/` before invoking executables; Stata and MATLAB preprocessing should be launched from the project files or batch files named in the relevant step.

Authors

- **Krzysztof Makarski** - SGH Warsaw School of Economics and FAME|GRAPE - kmakar@sgh.waw.pl
- **Joanna Tyrowicz** - University of Augsburg, FAME|GRAPE, University of Warsaw, and IZA - j.tyrowicz@grape.org.pl
- **Piotr Zoch** - University of Warsaw, FAME|GRAPE - p.zoch@grape.org.pl

Corresponding Author: Joanna Tyrowicz - j.tyrowicz@grape.org.pl

Replication Package Maintainer: Piotr Zoch - p.zoch@uw.edu.pl

License

The code in this repository is licensed under the MIT License.

See `LICENSE.txt` for the full license text.

Data licensing information is provided in the Data Availability section below.

Data Availability and Provenance Statements

Statement about Rights

The author(s) of the manuscript have legitimate access to and permission to use the data used in this manuscript.

The author(s) have documented permission to redistribute or publish the data contained within this replication package where redistribution is permitted. Data that should not be redistributed directly, including raw PSID and raw ACS/IPUMS extracts, is either omitted from the public package or should be shared only through provider-approved channels; the scripts and README document how replicators can obtain those inputs.

Summary of Availability

This paper uses public data, authors' estimates derived from restricted-use or terms-governed public-use microdata, and precomputed model inputs. The code in this repository is distributable under the MIT license; each data file remains subject to the license or terms of the original data provider.

All unrestricted processed inputs needed to run the Fortran scenarios are included. The PSID-derived income-process inputs used by the Fortran model (`_data_omega_*.txt` and `_data_sigma2eps_*.txt`) are included in `fortran_code/Data/`. When the replicator supplies the raw PSID extract, Step 2A refreshes the Stata-side components and Step 2B refreshes the MATLAB estimates.

The package includes the SCF+ file `data/SCF/SCF_plus.dta` (about 99 MB), which is used by the model-vs-data wealth-Gini and decomposition scripts. If it is missing from your copy, obtain SCF+ from the Kuhn-Scholarick-Steins data archive listed below and place it at that path.

Raw PSID microdata (Panel Study of Income Dynamics, 1970-2019 waves) should not be redistributed directly from a PSID website download. The authors' exact extract is archived in the official PSID Public Data Extract Repository (ICPSR ID 307887, <https://doi.org/10.3886/ICPSR307887.V1>), from which replicators can obtain the byte-identical `psid.dta` consistent with the PSID Conditions of Use; alternatively, rebuild it from <https://psidonline.isr.umich.edu/> using the shipped variable lists and `psid_read.R`. Raw PSID is not required to run the Fortran model with the shipped inputs. The package includes the processed PSID panel `data/PSID/psid_ready.dta` used by `outputs_stata_code/MvD_2_Gini_income.do` for Appendix Figure D.4; raw PSID is required only to regenerate that panel and the income-process pipeline from scratch.

The ACS/IPUMS extract is large (about 1.8 GB locally) and may be omitted from a public deposit if required by IPUMS terms. Despite its file name, `ACS_college.dta` **is raw data**: it is the IPUMS USA extract exactly as delivered by the IPUMS extract system - the authors only renamed the downloaded file; no observations or variables were modified. It contains the census/ACS samples for every year available at extraction time (decennial censuses 1940-2000, then ACS 2006, 2011, 2016, and 2021, reflecting the US move from a 10-year to a 5-year cadence). Note that IPUMS extracts are user-generated and re-downloading is not byte-reproducible; a fresh extract with the same samples and variables is equivalent in content but not bit-identical. The package includes the processed college-share inputs needed by the model; regenerating them from raw ACS requires `inputs_stata_code/skill_premium/ACS_college/ACS_college.dta`, obtained from IPUMS USA or redistributed only with the required IPUMS permission/journal-subset approval.

What Is Included vs. What Must Be Rebuilt

The package is organized so replicators can run the model from included processed inputs or rebuild those inputs from raw data where licensing permits. The information is presented as short records rather than a wide table so it remains readable in the PDF version.

PSID Income Process

Included: `fortran_code/Data/_data_omega_*.txt`, `fortran_code/Data/_data_sigma2eps_*.txt`, `data/PSID/psid_ready.dta`, and `data/PSID/ageeffects_*.dta`.

Raw input for a full rebuild: `inputs_stata_code/income_process/PSID/psid.dta`.

Transformation: Step 2A runs `inputs_stata_code/income_process/I01_run_psid_income_inputs.do` when the raw file is present. Step 2B then runs `inputs_matlab_code/income_process/estimate_parameters.m`.

Access and rights: Raw PSID is governed by the PSID Conditions of Use. Obtain the authors' exact extract from the PSID Public Data Extract Repository (<https://doi.org/10.3886/ICPSR307887.V1>) or rebuild it from the PSID Data Center. The exact extraction route is documented in `inputs_stata_code/income_process/PSID/psid_read.R` and the README in that directory.

ACS/IPUMS College Share

Included: `fortran_code/Data/_data_college_share.txt` and `inputs_stata_code/skill_premium/ACS_college/processed/col_share_acs*.dta`.

Raw input for a full rebuild: `inputs_stata_code/skill_premium/ACS_college/ACS_college.dta`.

Transformation: `inputs_stata_code/skill_premium/D02_prepare_college.do`.

Access and rights: A rebuild extract should use YEAR, AGE, HIGRADE, EDUCD, and PERWT for the years listed below. The IPUMS USA collection DOI is 10.18128/C010; cite the version DOI matching the extract date (Version 16.0, 10.18128/D010.V16.0, as of July 2026).

SCF+ Wealth Data

Included: `data/SCF/SCF_plus.dta`.

Transformation: The provider file is consumed directly by `outputs_stata_code/_prep_Gini_data.do` and `outputs_stata_code/MvD_3_GE_decomposition.do`; it is not produced by this package.

Source: Kuhn, Schularick, and Steins (2020).

Macro, Fiscal, Tax, and Aggregate Calibration

Included: The model-ready files in `fortran_code/Data/`, including `_data_depr.txt`, `_data_gamma.txt`, `_data_labsh.txt`, `_data_tau*.txt`, `_data_lambda.txt`, `_data_contrib_to_gdp.txt`, and `_data_skill_premium.txt`, plus the source snapshots used to create them.

Transformation: `inputs_stata_code/_main_data_prepare.do` and its component scripts. With `global download_data 0`, the scripts use packaged snapshots. Setting it to 1 refreshes supported dbnomics/OECD/GGDC series and overwrites those snapshots.

Mortality, Population, and Pension Replacement

Included: `fortran_code/Data/_data_pi_cond_US_since1935.txt`, `_data_het_pi_US_since1935_all.txt`, `_data_Nn_US_*.txt`, and `_data_rho_1935.txt`, together with the packaged HMD/UN/Case-Deaton/CDC/SSA source snapshots or hand-curated authority inputs.

Transformation: `demography/mortality/D01_life_tables.do`, `demography/hetero_pi/D03_prepare_hetero_pi.do`, and `external/copy_to_fortran.do`. The three files under `inputs_stata_code/external/` are copied into `fortran_code/Data/`; the mortality scripts document the HMD, UN, and Case-Deaton transformations.

No processed file used by the scripts is intended to be a manually edited black box. When a file is not regenerated by a script, it is listed above as either an external data product supplied by the original provider or a hand-curated authority input copied into place by `external/copy_to_fortran.do`.

Non-Proprietary Data Copies

Every Stata dataset shipped in this package is accompanied by a `.csv` copy with identical stored values (exported without value labels; variable and value labels remain documented in the `.dta` files). This includes the large redistributed files `data/SCF/SCF_plus.csv`, the processed PSID panel data `data/PSID/psid_ready.csv`, and all ten processed PSID intermediates under `inputs_stata_code/income_process/output/`. The only shipped `.dta` without a CSV copy is `data/model_psid_all_govt_.dta`, a regenerable model-to-Stata bridge that `outputs_stata_code/_main.do` rebuilds from `fortran_code/Results/`.

Raw Extract Specifications for Restricted Microdata

PSID: To rebuild the income-process files from scratch, place the raw public-use extract at `inputs_stata_code/income_process/PSID/psid.dta`. The authors' extraction uses the PSID Main Study family and individual files, the SRC sample, heads only, waves 1970-2019, and the `psidR` variable-list layout. The exact variable lists `famvars_big.txt` and

indvars.txt are included in inputs_stata_code/income_process/PSID/, and psid_read.R finds them there automatically (current psidR versions no longer ship famvars_big.txt; environment-variable overrides are documented in the local README). The first Stata stage then constructs household labor income, business-income components, household hours, average hourly income, CPI-deflated measures, education categories, birth cohorts, and cleaned age-20-to-65 person panels. The most important derived output for the figure pipeline is data/PSID/psid_ready.dta; it is produced by I01_02_make_variant_samples.do from the mostdrop_hhslabinc variant after dropping households whose five-year business-income share exceeds 25 percent and whose labor income is above the year-specific first quartile.

ACS/IPUMS: The authors' ACS_college.dta is the raw IPUMS USA extract as downloaded, without any modification (the extract file was only renamed). To rebuild the college-share input from scratch, create inputs_stata_code/skill_premium/ACS_college/ACS_college.dta from IPUMS USA (<https://usa.ipums.org/usa/>) using the samples/years 1940, 1950, 1960, 1970, 1980, 1990, 2000, 2006, 2011, 2016, and 2021 - every year available at the authors' extraction time. Required variables are YEAR, AGE, HIGRADE, EDUCD, and PERWT. IPUMS extracts are user-generated, so a re-download is content-equivalent but not byte-identical to the authors' file. Cite the IPUMS USA version DOI matching the extract date (collection DOI 10.18128/c010; Version 16.0 is 10.18128/D010.V16.0 as of July 2026); see inputs_stata_code/skill_premium/ACS_college/README.md. D02_prepare_college.do keeps age-25 observations, defines BA-or-higher status using HIGRADE before 1990 and EDUCD from 1990 onward, computes weighted shares, interpolates/extrapolates the five-year model path, writes fortran_code/Data/_data_college_share.txt, and saves the processed cohort files used by the heterogeneous-mortality script.

Model Input Catalogue

Each model input file, its source, and the script that produces it.

File	Source	Producer / location
_data_Nn_US_1935_2100.txt	UN World Population Prospects + CDC birth counts (https://population.un.org/wpp/ , https://www.cdc.gov/nchs/nvss/)	inputs_stata_code/external/ (hand-curved)
_data_Nn_US_1935_init_old.txt	CDC historical population tables	inputs_stata_code/external/ (hand-curved)
_data_rho_1935.txt	SSA PIA benefit formula (https://www.ssa.gov/oact/)	inputs_stata_code/external/ (hand-curved)
_data_pi_cond_US_since1935.txt	Human Mortality Database (https://www.mortality.org/) + UN WPP projections	demography/mortality/D01_life_tables.do
_data_het_pi_US_since1935_all.txt	Same as above, adjusted by education ratio from Case & Deaton (2021, PNAS 118(11)), shared by the authors	demography/hetero_pi/D03_prepare_hetero_pi.do
_data_omega_{mostdrop,busno_drop}_hhslabinc_avghourlyhh.txt	Derived from PSID 1970-2019 (https://psidonline.isr.umich.edu/)	inputs_stata_code/income_process/I01_run_psid_income_inputs.do
_data_sigma2eps_{mostdrop,busno_drop}_hhslabinc_avghourlyhh.txt	Derived from PSID 1970-2019	inputs_matlab_code/income_process/estimate_parameters.m
_data_gamma.txt	Penn World Table 10.0 rtfpna (https://www.rug.nl/ggdc/productivity/pwt/)	tfp/M02prepare_gamma.do
_data_gamma_robustness.txt	Same as above (alternative scenario path)	tfp/M02robustness_prepare_gamma.do
_data_tauC.txt, _data_tauK.txt, _data_tauL.txt	McDaniel (2007) updated tax series (https://drive.google.com/drive/folders/1O5ccfP2KN815y-OSp2hRMnHneW4IAkia)	tax_rate/T01prepare_taxes.do
_data_lambda.txt	Piketty-based progressivity via Bayer-Born-Luetticke replication data (tax_rate/progressivity_measures_all.xlsx)	tax_rate/T03prepare_tax_lambda.do
_data_contrib_to_gdp.txt	OECD Revenue Statistics (https://data-explorer.oecd.org/)	social_security/T02prepare_contribution.do
_data_depr.txt	Penn World Table 10.0 delta	depreciation/M01prepare_depr.do
_data_labsh.txt	Penn World Table 10.0 labsh	labor_share/M03prepare_labor_share.do

File	Source	Producer / location
_data_skill_premium.txt	Autor-Goldin-Katz (2020 AEA P&P), openICPSR 120694	skill_premium/H01prepare_skill_premium.do
_data_college_share.txt	ACS/decennial census microdata via IPUMS USA (https://usa.ipums.org/usa/); raw extract uses years 1940, 1950, 1960, 1970, 1980, 1990, 2000, 2006, 2011, 2016, and 2021 with YEAR, AGE, HIGRADE, EDUCD, and PERWT; processed inputs are shipped so the raw extract is needed only for a full rebuild	skill_premium/D02prepare_college.do
_data_exog_rate.txt	GGDC Penn World Table 10.0 irr.USA (internal rate of return); only consumed when switch _exog_rate=1 (e.g. the exor_ scenario, Figure F.7)	exog_rate/M04prepare_exog_rate.do

Auxiliary data consumed outside the Fortran input chain

- data/SCF/SCF_plus.dta - SCF+ harmonized wealth panel, Kuhn-Schularick-Steins (2020, *JPE*), about 99 MB, included in the package. If missing, obtain it from the supplementary material of Kuhn, Schularick, and Steins (2020) at <https://doi.org/10.1086/708815> and place it at this path. Consumed by outputs_stata_code/_prep_gini_data.do and MvD_3_GE_decomposition.do to produce the paper's wealth-Gini and decomposition figures.
- data/PSID/psid_ready.dta - processed PSID panel generated from the mostdrop income-process sample and used by outputs_stata_code/MvD_2_Gini_income.do for Appendix Figure D.4. The Stata PSID stage refreshes this file when raw PSID is available.
- inputs_stata_code/income_process/PSID/psid.dta - raw PSID public-use extract. Obtain the authors' exact extract from the PSID Public Data Extract Repository (ICPSR ID 307887, <https://doi.org/10.3886/ICPSR307887.V1>) or rebuild from <https://psidonline.isr.umich.edu/>. Consumed only if replicators re-estimate the income process from scratch or regenerate data/PSID/psid_ready.dta. Otherwise the pre-computed _data_omega_*.txt and _data_sigma2eps_*.txt files suffice for running the Fortran model.
- inputs_stata_code/skill_premium/ACS_college/processed/col_share_acs.dta and col_share_acs_ext.dta - processed ACS/IPUMS college-share inputs. The raw ACS_college.dta extract is needed only to rebuild these processed files. fortran_code/Data/_data_college_share.txt is already included for the Fortran model; see inputs_stata_code/skill_premium/ACS_college/README.md for exact IPUMS years and variables.

Data citations

Full data citations for every source, including access dates and DOIs, are collected in the References section at the end of this README; the manuscript carries the same citations.

Access, license, dates

All sources above are either public-domain, free for academic use under the provider's stated terms, or shipped here with the original authors' permission. Restricted or terms-governed raw microdata are redistributed only through the provider-approved channel (the PSID extract deposit) or omitted from the public package while retaining the processed files needed for replication. Source snapshots used in the paper were assembled during 2020-2023 and re-verified in 2026.

Computational Requirements

Software Requirements

Required Software

1. **Intel Fortran Compiler**
2. Version: Intel oneAPI Fortran Compiler ifx 2025.1.1, build 20250418 (the version used to build the shipped bin/5Gtrans.exe and bin/5Gtrans_het.exe)
3. Download: <https://www.intel.com/content/www/us/en/developer/tools/oneapi/fortran-compiler.html>
4. Note: The vfpjproj uses ifxCompiler for the x64 configurations (Release, Release_HetRate, Debug). The Win32 configurations declare ifortCompiler but are not used for the shipped binaries.
5. License: Free download available (may require registration)
6. **Microsoft Visual Studio**
7. Version: Visual Studio Community 2019, 16.11.55 (16.11.37206.5)
8. Components required:
 - C++ build tools (required by Fortran compiler)
 - Windows SDK
9. Download: <https://visualstudio.microsoft.com/downloads/>
10. License: Community Edition (free) is sufficient

11. **Stata** (for calibration inputs and paper figures)
 12. Version: Stata/SE 16.0
 13. Required packages: `mat2txt` (used by the income-process pipeline), `ineqdeco` (used by the Gini prep), and `lorenz` (used by Appendix Figure D.4). These are auto-installed on first run by the scripts that need them.
 14. Used by: `inputs_stata_code/` (Step 2, calibration inputs) and `outputs_stata_code/` (Step 7, paper figures)
 15. License: Commercial (StataCorp)
 16. **MATLAB**
 17. Version: MATLAB 9.5.0.944444, R2018b
 18. Used by: `inputs_matlab_code/income_process/` (Step 2B - income-process parameter estimation)
 19. License: Commercial (MathWorks)
 20. **Operating System**
 21. Windows 10 (64-bit) or Windows 11 (64-bit)
 22. Note: The batch scripts (.bat) and Visual Studio project files are Windows-specific
 23. **Linux/macOS compatibility:** The Fortran code itself is cross-platform compatible, but would require:
 - GNU Make or equivalent build system
 - Modification of the `system()` calls in `main.f90` (`mkdir/copy` commands)
 - Shell scripts to replace .bat files
-

Hardware Requirements

The Fortran solver allocates large arrays at startup, so memory is the binding constraint for the model runs. The requirements below reflect the configuration on which the authors actually ran the full pipeline; machines with substantially less memory may fail at startup with `fortr1: severe (41): insufficient virtual memory`, especially with a small or fixed-size Windows page file.

Verified Configuration (used for all shipped results)

- **Processor:** AMD Ryzen 7 5800X, 8 cores / 16 logical processors
- **RAM:** 96 GB
- **Storage:** 200 GB or more free space for the full `N_REPS=1000` rebuild. The Stata PSID bootstrap creates more than 110 GB of covariance files (roughly 55 MB per repetition for each of two variants), and the included Fortran result folders occupy about 27 GiB. A fast `N_REPS=0` or figure-only run needs substantially less space; allow at least 60 GB while extracting and working with the package.
- **Operating System:** Windows 11 (64-bit)

Notes for Smaller Machines

- The Stata and MATLAB preprocessing stages run on an ordinary 16 GB machine, but the full Stata bootstrap is disk-intensive: `inputs_stata_code/income_process/output/` can exceed 110 GB before MATLAB starts.
- We have not established a minimum RAM figure for the Fortran runs. On machines with 16 GB or less, let Windows manage the page file automatically (or set a large one) before attempting Step 4; a virtual-memory error at startup indicates the configuration is insufficient.
- Route 1 (figures from the included intermediate results) does not run the Fortran solver and works on an ordinary machine with enough disk space to extract the package.

Reference Hardware Environment

The authors' package verification and preprocessing runs used:

- **Operating system:** Microsoft Windows 11 Pro, version 10.0.26200, build 26200
- **Processor:** AMD Ryzen 7 5800X 8-Core Processor
- **Cores / logical processors:** 8 / 16
- **RAM:** 95.9 GB

Reference Software Environment

The authors' package verification used:

- **Stata:** Stata/SE 16.0 (C:\Program Files\Stata16\StataSE-64.exe)
- **MATLAB:** MATLAB 9.5.0.944444, R2018b (C:\Program Files\MATLAB\R2018b\bin\matlab.exe)
- **Fortran compiler:** Intel Fortran Compiler ifx 2025.1.1, build 20250418, via Intel oneAPI 2025.1
- **Visual Studio:** Visual Studio Community 2019, version 16.11.55 (16.11.37206.5), with C++ build tools

The shipped Fortran binaries are Windows x64 builds. Rebuilding from source should work with the compiler/toolchain above or with a compatible Intel oneAPI Fortran + Visual Studio installation.

Memory and Runtime Requirements

Summary

The Stata and MATLAB preprocessing stages are relatively short compared with the Fortran transition computations. The full set of paper scenarios is a multi-day job on a typical workstation, and individual scenarios can take hours depending on convergence speed and hardware.

The largest practical resource constraint is disk space, not the code itself. In the authors' complete results copy, `fortran_code/Results/` is about 27 GiB across 30 scenario folders. Most of this comes from `psid_all_govt__` (about 19.8 GiB) and `psid_ndm_govt__` (about 6.7 GiB).

Workload	Free disk space to allow
Figure-only check from the included intermediate results	60 GB
Baseline Fortran rerun without the full PSID bootstrap	80 GB
Full PSID bootstrap (N_REPS=1000) plus all included model results	200 GB minimum; 250 GB recommended

Scenario Runtime Guidance

Scenario group	Relative cost	Description
<code>psid_all_govt__</code>	High	Primary baseline; produces the largest downstream files used by most figures
<code>psid_n*</code>	High	Main counterfactuals for Figures 2-3 and appendices
<code>crr3_</code> , <code>ndel_</code> , <code>nstr_</code> , <code>gcbo_</code> , <code>beqs_</code>	High	Sensitivity families for Appendix F
<code>hrrat_</code>	High	Heterogeneous-return sensitivity family; run separately with <code>5Gtrans_het.exe</code>

Runtime Scaling

- Scenarios are independent and can be run in parallel if you launch separate processes, provided there is enough RAM and disk bandwidth.
- Release builds are required for practical runtimes; Debug builds can be much slower.
- Runtime varies with convergence speed, tolerance settings, compiler version, CPU speed, and whether antivirus or cloud-sync tools scan the output folder during runs.

Description of Programs and Code

Overview

This replication package contains a **computational general equilibrium model** of overlapping generations (OLG) with heterogeneous agents. The model is implemented in **Fortran 90** and uses **policy function iteration** to solve household optimization problems and **iterative methods** to find equilibrium prices and allocations.

Program Structure

Main Program Files

- `main.f90`: Entry point for the model
- Parses command-line arguments for scenario selection
- Sets up file paths and creates output directories
- Validates configuration files
- Allocates memory for large arrays
- Includes `main_base_transition.f90` for main computation
- `main_base_transition.f90`: Main computational logic (included in `main.f90`)
- Computes initial steady state (pre-reform)
- Computes final steady state (post-reform)
- Computes transition path between steady states

Core Computational Modules

- `steady_state.f90`: Steady state equilibrium solver
- Iterates over capital stock to find market-clearing equilibrium
- Calls household problem solver and aggregation routines

- Computes government budget balance and pension system balance
- **Key subroutine:** `steady()`
- `transition.f90`: Transition path solver
- Solves for perfect foresight transition between two steady states
- Handles cohort structure and time-varying parameters
- Iterates to convergence on price paths
- **Key subroutine:** `transition_path_DB()`

Household Problem Solvers

- `pfi_household_problem.f90`: Policy function iteration for household optimization
- Solves household Bellman equation via backward induction
- Handles 6-dimensional state space: age, assets, AIME, income shocks, return shocks, discount shocks
- Computes optimal consumption, labor supply, and savings
- **Contains:** `agent_vf()` and related optimization routines
- `pfi_aggregation.f90`: Aggregation across heterogeneous households
- Computes aggregate capital, labor, consumption
- Calculates distributional statistics (Gini coefficients, wealth shares)
- Aggregates by age, type, and across population
- **Key subroutine:** `get_aggregates()`
- `pfi_distribution.f90`: Distribution dynamics
- Computes stationary distribution of agents across states
- Forward simulation using policy functions
- Handles initial distribution and bequest receipts
- **Key subroutine:** `get_distribution()`
- `pfi.f90`: Master PFI module interface
- Defines state space grids and interpolation methods
- Sets up shock processes (income, return, discount)
- Provides utility functions and helper routines
- The file header documents the complete household problem formulation
- `pfi_het.f90`, `pfi_household_problem_het.f90`, `steady_state_het.f90`: heterogeneous-return variants of `pfi.f90`, `pfi_household_problem.f90`, and `steady_state.f90`; compiled only into `5Gtrans_het.exe` via the `Release_HetRate` configuration

Economic Model Components

- `pension_system.f90` and `pension_system_ss.f90`: PAYG pension system
- Calculates Social Security benefits using AIME formula
- Handles valorization and indexation of benefits
- Computes pension budget balance and subsidies
- Supports both defined benefit (PAYG) and defined contribution (funded) systems
- `ces_production.f90` and `ces_production_ss.f90`: Production function
- CES aggregation of heterogeneous labor types (college vs. non-college)
- Computes factor prices (wages, interest rates)
- Handles skill premium and substitution elasticity
- `cllosures.f90` and `closure_ss.f90`: Government budget
- Computes government revenues (taxes on labor, capital, consumption)
- Handles government debt dynamics
- Implements closure rule (government spending adjusts residually)
- `bequest.f90`: Bequest distribution
- Calculates accidental bequests from mortality
- Distributes bequests to surviving cohorts
- Supports equal distribution, pooling, or Zipf distribution

Data and Configuration

- `data.f90`: Data loading and processing
- Reads all external data files from `fortran_code/Data/` folder

- Processes demographics, taxes, productivity, mortality
- Extends time series to full transition horizon
- The file header lists all data files it reads
- **set_globals.f90**: Parameter initialization
- Reads configuration from `fortran_code/Instructions/` and `fortran_code/Parameters/` files
- Sets up model parameters, switches, and arrays
- Discretizes stochastic processes (AR(1) for shocks)
- The file header documents the initialization sequence
- **globals.f90**: Global variable declarations
- Defines all model parameters, arrays, and constants
- Dimensions: bigJ (ages), bigM (types), bigT (time periods)
- State space: n_a (assets), n_aime (AIME), n_sp (income), n_sr (return), n_sd (discount)

Utility Modules

- **AR_discrete.f90**: AR(1) process discretization
- Rouwenhorst method for discretizing autoregressive processes
- Used for income shocks, return shocks, discount factor heterogeneity
- **normalProb.f90**: Normal distribution utilities
- CDF and PDF calculations for normal distribution
- Used in shock process calibration
- **linint.f90**, **splines.f90**, **polynomial.f90**: Interpolation methods
- Linear, cubic spline, and polynomial interpolation
- Used for off-grid evaluation of value and policy functions
- **rootfinding.f90**, **minimization.f90**: Numerical optimization
- Root-finding for Euler equations
- Minimization for household optimization with constraints
- **matrixtools.f90**: Matrix operations
- Linear algebra routines
- Utilities for array manipulation
- **sort.f90**: Sorting algorithms
- Used for constructing wealth distribution
- Needed for Gini coefficient and percentile calculations
- **gini.f90**: Inequality measures
- Computes Gini coefficients
- Calculates wealth and income concentration

Output and Printing

- **Print_steady_DB.f90**: Steady state diagnostic output
- Prints equilibrium values, prices, aggregates
- Reports convergence diagnostics
- Included in `steady_state.f90` when `switch_print=1`
- **Print_DB.f90**: Transition path diagnostic output
- Prints time series of aggregate variables
- Reports feasibility and error metrics
- Included in `transition.f90`
- **print_iter.f90**: Iteration progress output
- Reports convergence progress during iterative solution
- Shows worst feasibility violations and errors
- **print_stamp.f90**: Timestamp utilities
- Date/time stamps for output files
- **pfi_print.f90**: Policy function diagnostics

- Prints policy functions for inspection/debugging

Support Files

- `Initial_values_DB.f90`: Initializes the transition-path arrays from the two steady states
- `get_profile_steady.f90`: Steady-state life-cycle profile computation (module `prof_steady`)
- `shocks_parameters.f90`: Shock-process parameter setup, included by `set_globals.f90`
- `transition_iterations.f90`: Household-problem step of the transition solver, included by `transition.f90`
- `clock.f90`: Timing utilities for performance measurement
- `assertions.f90`: Runtime assertion checking
- `errwarn.f90`: Error and warning message handling
- `gaussian_int.f90`: Gaussian quadrature for numerical integration
- `simplex.f90`: Simplex algorithm (not currently used in main path)
- `solver.f90`: Additional solver utilities

Build Configuration

- `5Gtrans.sln`: Visual Studio solution file
- `5Gtrans.vfproj`: Intel Fortran project file (contains compiler settings)
- Optimization: `/O2` (maximize speed)
- Precision: `/fp:precise`
- Runtime: `multithreaded (/threads)`

Batch Scripts

- `run_scenarios.bat`: Runs a predefined set of scenarios
- Hardcoded list of common scenarios
- Useful for quick testing
- `run_scenarios_from_list.bat`: Runs scenarios from `scenarios.txt`
- Reads scenario list from file (one per line)
- Flexible for custom scenario sets
- `scenarios.txt`: List of scenarios to run
- Format: `version experiment closure` (space-separated)
- Example: `psid_all_govt__`

Configuration Files

Instructions Files (`fortran_code/Instructions/` folder)

Files named: `{version}{experiment}{closure}instructions.txt`

Format: 37 lines, each containing an integer switch value followed by comment - Controls model features (mortality, taxes, pension rules, etc.) - Order of lines is CRITICAL and must match `set_globals.f90` read order

Example: `psid_all_govt__instructions.txt`

Parameters Files (`fortran_code/Parameters/` folder)

Files named: `{version}{experiment}{closure}parameters.txt`

Format: 49 lines containing numerical parameters (the two `hrat_` heterogeneous-return scenarios have 51: two extra lines at the end with the type-specific `rate_adj` values) - Tolerance levels, damping factors, structural parameters - Order of lines is CRITICAL and must match `set_globals.f90` read order

Example: `psid_all_govt__parameters.txt`

Output Files

Results are written to scenario-specific subfolders: `fortran_code/Results/{version}{experiment}{closure}/`

Output volume depends on three switches in the scenario's `instructions.txt`:

Switch	Baseline <code>psid_all_govt__</code>	All other shipped scenarios
<code>switch_print_macro</code> (line 36)	1	0
<code>switch_full_csv_write</code> (line 35)	1	0

Switch	Baseline psid_all_govt__	All other shipped scenarios
switch_small_write (line 33)	1	1

The baseline emits the full output set; every other scenario writes only the minimum needed to reproduce the paper figures (Gini series + steady-state summaries + run metadata). Older runs may have left additional *_j_trans.csv / mass_trans_minimal.csv files in non-baseline result folders - those are stale artifacts of earlier code and are **not** produced by the current build.

Current executable: always written by every fresh scenario run

- gini_trans.csv - Gini coefficient of savings by year (consumed by outputs_stata_code/_prep_Gini_data.do)
- steadys_old_information_run.txt - Initial steady-state summary (gated by switch_ss_write=1, true for all shipped scenarios)
- steadys_new_information_run.txt - Final steady-state summary (same gating)
- information.txt - Run configuration and parameter summary, written inside the scenario folder.
- implied_pi.txt - Mortality-transition diagnostic written inside the scenario folder.
- feasibility - Per-period feasibility check from the transition iterations
- {version}{experiment}{closure}instructions.txt, {version}{experiment}{closure}parameters.txt - Copies of the inputs used (auto-copied for reproducibility)

These statements define the output contract for a **fresh run with the current executable**. The 30 scenario folders supplied in fortran_code/Results/ are historical intermediate outputs generated before the current provenance-file placement and input-copy safeguards were introduced. Among those supplied folders, 7 contain information.txt, none contains implied_pi.txt, and none contains the copied instruction or parameter files. Their absence from a supplied historical folder does not indicate a failed run and does not prevent the figure-only route: the downstream Stata scripts use the included gini_trans.csv and other numerical model outputs. A fresh run should create all files in the list above inside its scenario folder.

Written only when switch_print_macro = 1 (baseline only)

Aggregate time series, one value per bigT period (1935-2100):

- Macro: gdp_trans.txt, gdp_pc_trans.txt, bigK_trans.txt, bigY_trans.txt, capital_trans.txt, y_trans.txt, cy_ratio_trans.txt, ky_ratio_trans.txt, ky_ratio_trans_1y.txt, iy_ratio_trans.txt, gamma_trans.txt, nu_trans.txt, Nt_trans.txt, lifeexp_trans.txt, depend_ratio_trans.txt, g_share_trans.txt, lambda_trans.txt, zet_trans.txt
- Rates and revenue: r_trans.txt, rate_trans.txt, rbar_trans.txt, r_pretax_trans_1y.txt, r_afterax_trans_1y.txt, irr_trans_1y.txt, r_low_trans.txt (only if rate_adj /= 0), r_type_trans.csv (same condition), tC_trans.txt, tL_trans.txt, tK_trans.txt, tC_tax_revenue_trans.txt, tL_tax_revenue_trans.txt, tK_tax_revenue_trans.txt
- Pension and debt: benefits_trans.txt, replacement_trans.txt, replacement2_trans.txt, contrib_to_gdp_trans.txt, subsidy_share_trans.txt, sum_b_weight_trans.txt, debt_share_trans.txt, debt_cost_share_trans.txt, savings_trans.txt, debt_trans.txt, b_scale_factor.txt, t1_additional_contrib.txt
- Welfare and superstars: u_init_old_trans.txt, u_all_trans.txt, u20_trans.txt, star_tinc_trans.txt, star_linc_trans.txt, star_pop_trans.txt, beq_gdp_trans.txt, avg_hours_trans.txt
- Distribution support: gini_weight_trans.txt, gini_weight_trans.csv

Written only when switch_full_csv_write = 1 (baseline only)

- prob_trans.csv - State-occupancy probabilities (year x age x asset x aime x income/return/discount shock)
- One of:
- mass_trans.csv (if switch_small_write = 0) - Full mass distribution with all variables (consumption, hours, income, wealth, savings)
- mass_trans_small.csv (if switch_small_write = 1) - Mass + pretax labor income + savings only (this is what the baseline scenario writes)
- mass_trans_beq.csv - Bequest-distribution mass (only if also switch_unequal_bequest = 2; it is not produced by the baseline and is not used by any shipped figure)

Fresh-Run Output Note

The R2 reproducibility check found two files in a fresh baseline run that were absent from the authors' older intermediate folder, and seven files in that older folder that a fresh run did not produce. This is expected after the output cleanup in the current build:

- implied_pi.txt and information.txt are current provenance outputs. Older builds could write them outside the scenario folder; the current build writes and replaces both inside fortran_code/Results/<scenario>/.
- b_j_trans.csv, c_j_trans.csv, l_j_trans.csv, sv_j_trans.csv, and the baseline mass_trans_beq.csv were empty legacy placeholders.
- fort.30 was an implicit-unit legacy output, and u_j_trans.csv was an obsolete diagnostic.

None of these seven historical files is read by a downstream Stata, MATLAB, or Fortran stage. Six names (b_j_trans.csv, c_j_trans.csv, l_j_trans.csv, sv_j_trans.csv, fort.30, and u_j_trans.csv) are no longer written by the current source. mass_trans_beq.csv is the exception: it remains a valid conditional output when switch_full_csv_write = 1 and switch_unequal_bequest = 2, but the baseline does not satisfy that condition. The zero-byte historical placeholders have been

removed from the included results. A scenario run clears all seven names at startup so stale artifacts cannot survive an in-place rerun; when the bequest-output condition is met, the current code subsequently recreates `mass_trans_beq.csv`.

`switch_full_csv_write = 2` (**compact mode, not used by any shipped scenario**)

If you set this manually, you'll get `mass_trans_medium.csv` (`switch_small_write=0`) or `mass_trans_minimal.csv` (`switch_small_write=1`) instead of the `=1` outputs above.

License for Code

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Instructions to Replicators

Prerequisites

Before beginning a full replication from source, ensure you have: 1. Installed Intel Fortran Compiler (see Software Requirements) 2. Installed Visual Studio with C++ build tools (see Software Requirements) 3. At least 60 GB free disk space for a figure-only check, 80 GB for a baseline rerun, or 200 GB for the full `N_REPS=1000` pipeline (see Hardware Requirements)

For the plot-only fast path below, the Fortran compiler and Visual Studio are not needed if the precomputed `fortran_code/Results/` folders are already present.

Step-by-Step Replication Instructions

Each step below is organized as **Inputs**, **Run**, **Outputs**, and **Details**. This is meant to separate what the replicator must supply from what the code produces.

Step 1: Obtain and Check the Package

Inputs

- The extracted `3-replication-package` folder.
- Optional restricted raw-data files only if doing Route 3.

Run

Download and extract the package to any local path with enough disk space. No hardcoded locations are assumed.

```
dir
dir fortran_code\Results
```

Outputs

- A local working copy with this README, source code, shipped inputs, and the included `fortran_code/Results/<scenario>/` folders.

Details

The intermediate Fortran result folders are included inside the main replication package at `fortran_code/Results/`; there is no separate companion results archive. After extraction, one folder per scenario should appear there. The two largest folders are `psid_all_govt__` and `psid_ndm_govt__`. The baseline folder contains the large `mass_trans_small.csv` and `prob_trans.csv` files used by the Stata figure pipeline.

Step 2: Generate Calibration Inputs

The Fortran model reads calibration files from `fortran_code/Data/`. Step 2 is split into a Stata stage for macro, demographic, fiscal, ACS, and PSID covariance inputs, and a MATLAB stage for the PSID income-process parameters.

Inputs

Input	Required for	Location / setting
Packaged source snapshots	Routes 1-3	<code>inputs_stata_code/</code> , <code>data/</code> , and <code>fortran_code/Data/</code>
Raw PSID extract	Route 3 PSID rebuild only	<code>inputs_stata_code/income_process/PSID/psid.dta</code>
Raw ACS/IPUMS extract	Route 3 ACS rebuild only	<code>inputs_stata_code/skill_premium/ACS_college/ACS_college.dta</code>
Stata	Step 2A, including the conditional PSID stage	Stata 16+; packages are auto-installed where needed
MATLAB	MATLAB stage of Step 2B and Appendix Figure B.5	MATLAB_EXE can override the default executable path
Bootstrap setting	Step 2B	<code>inputs_stata_code/income_process/N_REPS.txt</code>

Run

Step 2A, the general Stata calibration stage:

1. Open `inputs_stata_code/main.stpr` in interactive Stata.
2. Open `__main_data_prepare.do` inside that project.
3. Run the do-file.

Step 2B, the MATLAB income-process stage, only if re-estimating PSID-derived parameters:

Step 2A has already run `income_process/I01_run_psid_income_inputs.do` when raw `PSID/psid.dta` is present. Do not run that do-file again. From the repository root, run:

```
set "MATLAB_EXE=C:\Program Files\MATLAB\R2018b\bin\matlab.exe"
inputs_matlab_code\income_process\run_income_process_matlab.bat
```

For Appendix Figure B.5 only, without re-estimating parameters, run:

```
"C:\Program Files\MATLAB\R2018b\bin\matlab.exe" -batch "run('outputs_matlab_code/income_process/plot_estimates.m')"
```

Outputs

Output class	Files written or refreshed
Fortran calibration inputs	<code>fortran_code/Data/_data_*.txt</code>
PSID processed panels	<code>data/PSID/psid_ready.dta</code> , <code>data/PSID/ageeffects_*.dta</code> when raw PSID is supplied
ACS processed panels	<code>inputs_stata_code/skill_premium/ACS_college/processed/col_share_acs*.dta</code> when raw ACS/IPUMS is supplied
Calibration figures	<code>graphs/inputs/*.{png,eps,gph,svg,pdf}</code> for Stata figures; <code>graphs/inputs/sigma2eps_*.{eps,png}</code> for MATLAB
MATLAB live outputs	<code>inputs_matlab_code/income_process/output/<variant>/{H,L}.mat</code>

Details

The Stata drivers use project-relative paths. Open them from `inputs_stata_code/main.stpr`; otherwise references such as `%source/bone` and `../fortran_code/Data/_data_$var.txt` may resolve incorrectly.

The `global download_data` switch is set to 0 at the top of the relevant Stata drivers. With the default value, scripts use packaged source snapshots and do not require network access. Setting it to 1 refreshes supported dbnomics/OECD/GGDC series and overwrites the local snapshots.

Raw ACS/IPUMS and raw PSID are conditional. If `ACS_college.dta` is absent, Step 2A keeps the packaged college-share inputs. If `psid.dta` is present, Step 2A runs `I01_run_psid_income_inputs.do` exactly once; if it is absent, Step 2A keeps the packaged PSID-derived Fortran inputs and processed `psid_ready.dta`. Run the PSID do-file directly only as a standalone alternative when deliberately skipping the rest of Step 2A. The exact raw-data specifications are in the Data Availability section and the local README files under `inputs_stata_code/income_process/PSID/` and `inputs_stata_code/skill_premium/ACS_college/`.

The full catalogue of `_data_*.txt` inputs and their producing scripts is in the Model Input Catalogue above. Three Fortran inputs have no Stata producer and are copied from `inputs_stata_code/external/` by `external/copy_to_fortran.do`: `_data_Nn_US_1935_2100.txt`, `_data_Nn_US_1935_init_old.txt`, and `_data_rho_1935.txt`.

The shared bootstrap setting is `inputs_stata_code/income_process/N_REPS.txt`. The shipped default is 1000, which reproduces the paper's full bootstrap when you run both the Stata PSID stage and the MATLAB stage: expect about 12 hours in total, roughly 1 hour for Stata covariance generation plus roughly 11 hours for MATLAB `lsqnonlin` bootstrap estimation. For a fast run that computes point estimates only and reuses the archived 1000-repetition bootstrap files for the paper confidence bands, set that file to:

0

MATLAB checks that the matching Stata bootstrap covariance files exist before starting a bootstrap run; if they are missing, it stops with an explicit missing-file message rather than a low-level `textscan` error.

`inputs_matlab_code/income_process/output/<variant>/{H,L}.mat` are live outputs and are overwritten by the MATLAB stage. The companion `{H,L}_archive.mat` files hold the authors' paper-baseline 1000-repetition bootstrap arrays.

`outputs_matlab_code/income_process/plot_estimates.m` uses live point estimates plus archived confidence bands, so a fast `N_REPS=0` run does not erase the paper confidence intervals.

To verify Step 2, check:

```
dir fortran_code\Data
dir graphs\inputs
```

Step 3: Compile the Model or Use Shipped Binaries

Inputs

- `fortran_code/*.f90`, `fortran_code/5Gtrans.sln`, and `fortran_code/5Gtrans.vfproj` if rebuilding.
- Intel oneAPI Fortran and Visual Studio with C++ build tools if rebuilding.
- `fortran_code/bin/5Gtrans.exe` and `fortran_code/bin/5Gtrans_het.exe` if using shipped binaries.

Run

Option 0, use shipped binaries:

```
dir fortran_code\bin
```

Rebuild with the supplied Visual Studio solution:

1. Open `fortran_code/5Gtrans.sln`.
2. Select Release and x64.
3. Build > Rebuild Solution.
4. Select Release_HetRate and x64.
5. Build > Rebuild Solution again for the heterogeneous-return executable.

The two configurations deliberately select mutually exclusive implementations: Release uses `pfi.f90` and `steady_state.f90`, while Release_HetRate uses `pfi_het.f90` and `steady_state_het.f90`. Do not compile with a wildcard such as `ifx *.f90`; that selects both implementations of the same modules and does not preserve the project dependency order. The supplied solution and project files are the supported source-rebuild route.

Outputs

- Main build: `fortran_code/x64/Release/5Gtrans.exe`.
- Heterogeneous-return build: `fortran_code/x64/Release_HetRate/5Gtrans_het.exe`.
- Shipped binary locations: `fortran_code/bin/5Gtrans.exe` and `fortran_code/bin/5Gtrans_het.exe`.

Details

The Release_HetRate configuration is needed for the `hrat_` sensitivity scenarios in Appendix F. It swaps in the heterogeneous-return source files through the Visual Studio project configuration. Compilation typically takes 2-5 minutes.

Step 4: Run a Single Scenario Test

Inputs

- A main Fortran executable from Step 3.
- Calibration inputs in `fortran_code/Data/`.
- Matching instruction and parameter files in `fortran_code/Instructions/` and `fortran_code/Parameters/`.

Run

From `fortran_code/`, run the baseline scenario:

```
cd fortran_code
bin\5Gtrans.exe psid_all_govt__
```

After a local Visual Studio rebuild, you can instead run:

```
x64\Release\5Gtrans.exe psid_all_govt__
```

Outputs

- `fortran_code/Results/psid_all_govt__/`.
- Steady-state summaries, convergence diagnostics, aggregate time series, distribution files, and `gini_trans.csv`.

Details

This is a long run; expect hours for the baseline scenario on a typical workstation. The program prints the scenario name, then initial steady-state progress, final steady-state progress, and transition-path progress. Convergence is reached when the model's error criterion falls below the tolerance in the parameter file.

Step 5: Verify Scenario Output Folders

Inputs

- `fortran_code/Results/psid_all_govt__/` or another completed scenario folder.

Run

```
dir fortran_code\Results\psid_all_govt__
```

Outputs to confirm after the fresh Step 4 run

- `steadys_old_information_run.txt` and `steadys_new_information_run.txt`.
- `information.txt` and `implied_pi.txt`.
- `gini_trans.csv`, consumed by downstream Stata scripts.
- Baseline aggregate series such as `gdp_trans.txt`, `r_trans.txt`, `bigK_trans.txt`, and `bigY_trans.txt`.
- Baseline full-output files `mass_trans_small.csv` and `prob_trans.csv`.
- Copied `psid_all_govt__instructions.txt` and `psid_all_govt__parameters.txt`.

Details

The baseline scenario writes the large output set because it sets `switch_print_macro=1` and `switch_full_csv_write=1`. Most non-baseline scenarios write a smaller set, primarily `gini_trans.csv`, steady-state summaries, and run metadata. The full switch-by-switch output list is in the Output Files section.

For the **supplied historical folders**, confirm the numerical files needed by the downstream route, especially `gini_trans.csv`, the two steady-state summaries, and the baseline aggregate and distribution files listed above. Do not require `information.txt`, `implied_pi.txt`, or copied instruction/parameter files in those supplied folders: they predate the current provenance safeguards. The Economic Journal reproducibility team confirmed the tested baseline scenario starting from raw data and then rebuilt all paper results from the intermediate outputs supplied in this package.

Step 6: Run All Model Scenarios

Inputs

- Completed Step 2 calibration inputs.
- Main executable `5Gtrans.exe`.
- Heterogeneous-return executable `5Gtrans_het.exe` for `hrat_` scenarios.
- Scenario definitions in `fortran_code/scenarios.txt`, `fortran_code/Instructions/`, and `fortran_code/Parameters/`.

Run

From `fortran_code/`:

```
cd fortran_code
run_scenarios_from_list.bat
run_scenarios_het.bat
```

Outputs

- One folder per completed scenario under `fortran_code/Results/`.
- The result folders consumed by Step 7 and listed in `SCENARIOS.md`.

Details

`run_scenarios_from_list.bat` reads every uncommented scenario in `scenarios.txt` and runs them sequentially with `5Gtrans.exe`. It searches the current folder, `x64\Release\`, `x64\Debug\`, `Release\`, `Debug\`, and then `bin\` as a shipped-binary fallback.

The `hrat_` scenarios are not in `scenarios.txt`; run them with `run_scenarios_het.bat`, which invokes `5Gtrans_het.exe`. The full paper mapping from scenario family to figure is in `SCENARIOS.md`. The complete scenario set is a multi-day workload.

To save a console log:

```
run_scenarios_from_list.bat > replication_log.txt 2>&1
```

Scenarios are independent and can be run in parallel in separate command windows if the machine has enough RAM and disk bandwidth.

Step 7: Generate Paper Tables and Figures

Inputs

- Scenario folders under `fortran_code/Results/`.
- `data/SCF/SCF_plus.dta` for wealth model-vs-data figures.
- `data/PSID/psid_ready.dta` for Appendix Figure D.4.
- Stata project outputs `_stata_code/___replication_graphs.stpr`.

Run

1. Open `outputs_stata_code/___replication_graphs.stpr` in interactive Stata.
2. Open `__main.do` inside the project.
3. Run the do-file.

If the complete result folders are already present, Step 7 is the Route 1 fast path and Steps 2-6 can be skipped.

Outputs

- Main and appendix figures in `graphs/outputs/`.
- Appendix calibration/input figures in `graphs/inputs/`.
- Intermediate Stata datasets in `data/` where required by the plotting pipeline.

Details

`outputs_stata_code/___main.do` sets:

- `$resultspath = "..\fortran_code\Results\"`
- `$graphspath = "..\graphs\outputs\"`
- `$datapath = "..\data\"`

The driver produces main-text figures, Appendix B calibration figures, Appendix C population-structure figures, Appendix D model-vs-data figures, Appendix E decompositions, and Appendix F sensitivity figures. The full figure-to-script mapping is in the List of Tables and Figures section.

Appendix Figure B.1, the college-share calibration figure, is rebuilt only when `inputs_stata_code/skill_premium/ACS_college/ACS_college.dta` is present. If the raw ACS/IPUMS extract is omitted, the driver prints a message and keeps the packaged processed college-share inputs and existing `graphs/inputs/college_share.*` files.

`__main.do` assumes that every scenario referenced by the requested figures has already been run. Missing scenarios will surface as Stata errors pointing to the specific `.csv` or result folder that could not be imported.

The Stata drivers create transient `bone.dta` and `bone1.dta` scratch files in `outputs_stata_code/`. They are regenerated on every run, read-only attributes are cleared before overwrite/delete attempts, and the files are safe to delete between runs.

Verification

Comparing Results to Paper

Use the replication crosswalk near the end of this README to map each paper table and figure to the Stata driver, Fortran scenario family, and output files that generate it. For the main numerical series, compare the files in `fortran_code/Results/psid_all_govt___/` and the generated figures in `graphs/outputs/` against the paper.

Numerical Precision

Bit-for-bit reproduction is not guaranteed across compilers and hardware. As a reference point, the Economic Journal reproducibility team's independent rerun of the baseline scenario `psid_all_govt___` matched the supplied intermediate outputs to approximately $1e-06$.

Small differences may occur due to: - Different compiler versions (Intel Fortran Classic vs. oneAPI) - Different optimization flags - Different operating system versions - Hardware floating-point rounding differences

Large differences (> 1%) indicate a problem: - Check that you're using the correct scenario - Verify all data files are present and unchanged - Check compiler settings (must be Release mode with /O2) - Verify instructions and parameters files match the provided versions

Troubleshooting

Compilation Errors

Error: "Cannot open module file" or "Error opening compiled module" - **Solution:** Clean the solution (Build > Clean) and rebuild - Delete all `.mod` and `.obj` files in `x64/Release/` - Rebuild from scratch

Error: "Link error: unresolved external" - **Solution:** Ensure all `.f90` files are included in the project - Right-click project > Add > Existing Item > Select missing `.f90` files

Runtime Errors

Error: "Configuration file not found" - **Solution:** Check that `fortran_code/Instructions/` and `fortran_code/Parameters/` folders contain the required files - Verify file names exactly match: `{version}{experiment}{closure}instructions.txt`

Error: "Stack overflow" or "Access violation" - **Solution:** Increase stack size in project properties - Project Properties > Fortran > Optimization > Stack Size > Set to "unlimited" or large value (e.g., 500000000)

Error: Program hangs (no progress for hours) - **Solution:** May be stuck in convergence loop - Check parameters file: ensure tolerance (`err_ss_tol`) is not too small (e.g., use $1e-7$, not $1e-12$) - Check damping parameter (`up_ss`) is between 0.3-0.8 (too high causes oscillation, too low is slow)

Incorrect Results

Issue: Results differ significantly from paper 1. Verify scenario name is correct 2. Check that instructions and parameters files haven't been modified 3. Ensure Release mode (not Debug) was used 4. Verify compiler optimization is enabled (`/O2`) 5. Check data files haven't been corrupted (compare checksums if provided)

Performance Issues

Issue: Runs much slower than expected runtime - Check that Release mode is used (Debug is 10-50x slower) - Verify CPU is not throttling (check temperatures, power settings) - Close other applications (browser, etc.) to free RAM - Check that antivirus isn't scanning the output folder during run

Getting Help

`SCENARIOS.md` contains the full per-scenario catalogue, including which paper figure each scenario supports. For questions about replication, contact the package maintainer listed at the top of this README.

List of Tables and Figures

This section maps each numbered figure in the paper to the output files and programs that generate it. The paper contains no numbered tables.

Reproducibility Status

The provided code and shipped inputs reproduce the paper conditional on the data inputs listed above: - All numbers provided in text in the paper - All main-text figures - All scripted online-appendix figures - All scripted online-appendix input-series figures

Data caveat: regenerating the PSID-derived income-process files requires the raw PSID extract; regenerating the ACS college-share input requires the raw IPUMS extract. The package includes processed versions needed for the Fortran model and Appendix Figures B.4 and D.4, plus SCF+ for the model-vs-data wealth figures.

Main Text

Figure	Paper title	Generating script	Output file
Figure 1	Evolution of wealth inequality: model vs. data	<code>outputs_stata_code/R_Figure1.do</code>	<code>graphs/outputs/Results_Gini_changes.png</code>
Figure 2	Baseline vs counterfactual scenario of constant longevity	<code>outputs_stata_code/R_Figure2.do</code>	<code>graphs/outputs/Results_Gini_counterfactuals.png</code>
Figure 3	Impact of rising longevity on wealth inequality (in Gini points)	<code>outputs_stata_code/R_Figure3.do</code>	<code>graphs/outputs/Results_Gini_drivers_demographics.png</code>
Figure 4	Impact of different factors evolution on wealth inequality (in Gini points)	<code>outputs_stata_code/R_Figure3.do</code>	<code>graphs/outputs/Results_Gini_drivers_comparison.png</code>

The paper's main text contains no tables.

Online Appendix B - Calibration

Figure	Paper title	Generating script	Output file(s)
Figure B.1	Share of college graduates	<code>skill_premium/D02_prepare_college.do</code>	<code>graphs/inputs/college_share.*</code>
Figure B.2	Life expectancy at 50 (college vs. non-college)	<code>demography/hetero_pi/D03_prepare_hetero_pi.do</code>	<code>graphs/inputs/LE50year.*</code>

Figure	Paper title	Generating script	Output file(s)
Figure B.3	Skill premium	skill_premium/H01prepare_skill_premium.do	graphs/inputs/skill_premium.*
Figure B.4	Deterministic profile of log productivity across age	outputs_stata_code/income_process/M05plot_omega.do	graphs/inputs/omega_deaton_mostdrop_hhslabinc.*
Figure B.5	Variances of idiosyncratic productivity shocks	outputs_matlab_code/income_process/plot_estimates.m	graphs/inputs/sigma2eps_{mostdrop,busno_drop}_hhslabinc.{eps,png}
Figure B.6	Technology - panels (a) depreciation, (b) labor share, (c) TFP	depreciation/M01prepare_depr.do; labor_share/M03prepare_labor_share.do; tfp/M02prepare_gamma.do	graphs/inputs/depr.*, lab_share.*, gamma.*
Figure B.7	Tax rates - panels (a-c) tau_C/tau_K/tau_L, (d) progressivity lambda	tax_rate/T01prepare_taxes.do; tax_rate/T03prepare_tax_lambda.do	graphs/inputs/tC.*, tK.*, tL.*, lambda.*
Figure B.8	Social security contributions to GDP	social_security/T02prepare_contributions.do	graphs/inputs/contributions.*
Figure B.9	Replacement-rate scale parameter rho_j,t	inputs_stata_code/social_security/T04plot_rho.do (invoked from outputs_stata_code/__main.do)	graphs/inputs/rho.*

Online Appendix C - Population structure

Figure	Paper title	Generating script	Output file(s)
Figure C.1	Population structure: baseline vs. frozen-1955-longevity counterfactual, seven periods (1935, 1950, 1975, 2000, 2020, 2050, 2100)	outputs_stata_code/R_FigureC1_popstructure.do (reads population.csv from both psid_all_govt_/ and psid_ndm_govt_/ Results folders)	graphs/outputs/AppC_PopStructure_{1935,1950,1975,2000,2020,2050,2100}.{png,svg}

Online Appendix D - Model vs. data

Figure	Paper title	Generating script	Output file
Figure D.1	The real interest rate, model vs data	outputs_stata_code/MvD_1_macro.do	graphs/outputs/irr_trans_levels.*
Figure D.2	Average annual hours per capita aged 20-64, model vs data	outputs_stata_code/MvD_1_macro.do	graphs/outputs/avghours_trans_levels.*
Figure D.3	Share of expenditure on social security benefits in GDP, model vs data	outputs_stata_code/MvD_1_macro.do	graphs/outputs/benefits_trans_levels.*
Figure D.4	Lorenz curves of income distribution (five decade snapshots, 1970-2010)	outputs_stata_code/MvD_2_Gini_income.do	graphs/outputs/Lorenz_{1970,1980,1990,2000,2010}.*
Figure D.5	Contribution of between-cohort vs. within-cohort inequality	outputs_stata_code/MvD_3_GE_decomposition.do	graphs/outputs/MvD_GE.*

Online Appendix E - Additional decompositions

Figure	Paper title	Generating script	Output file
Figure E.1	Impact of evolution of income determinants on wealth inequality	outputs_stata_code/R_Figure3.do (psid_ncs/_ncp/_nsh_variants)	graphs/outputs/Results_Gini_drivers_incomes.png
Figure E.2	Impact of tax changes on wealth inequality	outputs_stata_code/R_Figure3.do (psid_ntl/_ntc/_ntk/_ntp_variants)	graphs/outputs/Results_Gini_drivers_taxes.png
Figure E.3	Impact of technology changes on wealth inequality	outputs_stata_code/R_Figure3.do (psid_nls/_nga/_ndp_variants)	graphs/outputs/Results_Gini_drivers_macro.png

Online Appendix F - Sensitivity

Figure	Paper title	Generating script	Output file
Figure F.1	Baseline vs. constant-longevity (theta = 3)	outputs_stata_code/R_Figure2.do (crr3_variants)	graphs/outputs/AppF_Gini_counterfactuals_theta.png

Figure	Paper title	Generating script	Output file
Figure F.2	Baseline vs. constant-longevity (heterogeneous returns)	outputs_stata_code/R_Figure2.do (hrat_ variants)	graphs/outputs/AppF_Gini_counterfactuals_hetrates.png
Figure F.3	Baseline vs. constant-longevity (no discount-factor shocks)	outputs_stata_code/R_Figure2.do (ndel_ variants)	graphs/outputs/AppF_Gini_counterfactuals_homogendelta.png
Figure F.4	Baseline vs. constant-longevity (no "superstars")	outputs_stata_code/R_Figure2.do (nstr_ variants)	graphs/outputs/AppF_Gini_counterfactuals_nosuperstars.png
Figure F.5	Primary and alternative TFP growth rate	tfp/M02robustness_prepare_gamma.do	graphs/inputs/gamma_robust.*
Figure F.6	Baseline vs. constant-longevity (TFP growth rate)	outputs_stata_code/R_Figure2.do (gcbo_ variants)	graphs/outputs/AppF_Gini_counterfactuals_gamma.png
Figure F.7	Baseline vs. open economy (exogenous interest rate)	outputs_stata_code/R_Figure1_ap.p.do (exor_ variant)	graphs/outputs/AppF_Gini_counterfactuals_exograte.png
Figure F.8	Baseline vs. constant-longevity (unequal bequest distribution)	outputs_stata_code/R_Figure2.do (beqs_ variants)	graphs/outputs/AppF_Gini_counterfactuals_beq.png

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Corresponding Author: Joanna Tyrowicz - j.tyrowicz@grape.org.pl